

ReactiveLJ: A Coordination-Dependent Lennard–Jones Potential for Reversible Bonding in Coarse-Grained Polymer Simulations

Matthew D. Chertok and Michael A. Webb*

Department of Chemical and Biological Engineering, Princeton University, Princeton, NJ 08544, USA

E-mail: mawebb@princeton.edu

Abstract

Associative polymer networks are topologically adaptive materials whose reversible bonds shape their structure, transport, and viscoelastic response. Molecular simulation offers a way to connect microscopic bond kinetics with macroscopic rheology, but existing approaches often rely on prescribed Monte Carlo moves or complex many-body reactive potentials that require substantial parameter tuning or multiple reactive species. Here, we introduce ReactiveLJ, a phenomenological coordination-dependent Lennard-Jones interaction for reversible bonding between coarse-grained reactive stickers. The model employs a smooth local coordination measure to weaken attraction under crowding and incorporates force-continuous smoothing to maintain stable dynamics across the repulsive-attractive crossover. Since bond formation and exchange arise directly from the potential energy landscape, the interaction remains well-defined under nonequilibrium conditions. We use the model to study associative polymer melts across a range of cohesive strengths and characterize network self-assembly and gelation, linear viscoelasticity through the relaxation modulus $G(t)$, and the resulting bond-exchange

kinetics. Overall, ReactiveLJ provides a simple and efficient reactive coarse-grained potential for modeling associative polymer networks in and out of equilibrium.

1 Introduction

Associative polymer networks are materials that can adapt their topology through the formation of reversible bonds between polymer chains.^{1,2} These interactions are fundamental to many biological self-assembly processes and have influenced the design of synthetic polymers that use reversible bonding,³ such as nylon⁴ and Kevlar.⁵ In synthetic polymers, hydrogen bonding, metal-ligand coordination, ionic association, and dynamic covalent exchange have been used to impart adaptive functions such as self-healing and thermal reshaping, and supramolecular sticker concepts can also be applied to tune adhesive performance.^{1,2,6–8} By substituting permanent covalent cross-links with exchangeable bonds, networks can be created that relieve stress and can be reshaped or recycled under appropriate conditions.^{2,6}

To connect microscopic association and exchange to macroscopic response, mean-field and scaling theories for reversible gels and transient networks have been developed, providing a framework for linear viscoelasticity under equilibrium conditions.^{9,10} These theories—including reversible-gelation ideas and the sticky Rouse/sticky reptation description of associating polymers—often provide a useful baseline for interpreting equilibrium linear viscoelasticity when stickers are well spaced along the chains, the network is sufficiently uniform that fluctuations can be handled by controlled averaging, and bond-exchange kinetics are well separated from the underlying chain relaxation dynamics.^{1,3,9,10} In contrast, MD resolves the coupled conformational and bonding dynamics by numerically integrating the equations of motion for many interacting degrees of freedom. For example, the linear viscoelastic response can be obtained from equilibrium stress autocorrelation functions via Green-Kubo relations.¹¹ However, for phenomenological models such as Kremer-Grest polymers,¹² an important modeling decision is how to represent bond formation, breakage, and exchange

at the coarse-grained level. Prior studies have implemented reversible bonding through hybrid molecular dynamics (MD) and Monte Carlo (MC) moves, or related stochastic reaction rules.^{13,14} These schemes can efficiently sample network topology and encode chemistry-inspired reaction pathways, but they typically require user-prescribed rates and/or external acceptance criteria rather than kinetics that emerge solely from the interaction potential.^{3,14}

An alternative is to embed reactivity into the potential energy landscape so that association and exchange arise directly from forces and thermal fluctuations during standard MD.^{15,16} Minimal reactive constructions based on coordination-dependent many-body terms show that associative bond swaps can be realized without topology-altering MC moves by introducing a tunable exchange barrier controlled by a single parameter.^{15,16} In practice, these three-body swap potentials are designed to enforce restricted bonding rules, most notably single-bond-per-site constraints, and are therefore most natural for comparatively structured association motifs.^{15,16} Extending the approach to less structured bonding patterns, such as associative polymer melts, requires additional modeling flexibility in how association rules and competing interactions are encoded.^{15,16}

Reactive bond-order force fields offer a practical approach to depict bonding and exchange kinetics as outcomes of a unified energy landscape. In coarse-grained, coordination-dependent models, the Tersoff functional form is a widely used choice for implementing this idea.¹⁷ By using environment-dependent, many-body interaction energies, this potential can model both bond formation and bond breaking within a single Hamiltonian description.¹⁷ In line with this approach, Liu and O'Connor showed that a tailored Tersoff bond-order potential can be applied in a coarse-grained bead-spring melt to model associative polymer networks in both equilibrium and non-equilibrium states, with bond kinetics and exchange pathways naturally arising from the reactive landscape.³ Their work reveals that a Hamiltonian reactive model can describe dynamics under non-equilibrium flow conditions, unlike MC approaches which impose kinetics through equilibrium-based rules.³

However, the Tersoff potential contains coordination terms that depend on both local

crowding and angular contributions, which were initially developed for modeling covalent materials.¹⁷ By contrast, much of coarse-grained polymer physics relies on the Kremer-Grest family of bead-spring models,¹² which use truncated and shifted Lennard–Jones (LJ) interactions or their purely repulsive (WCA) variants. When the desired associative interactions are effectively isotropic at the chosen level of detail, explicit angular terms become unnecessary and expensive.¹⁸ Furthermore, the increased functional complexity and larger parameter set can make tuning challenging when the main goal is to describe crowding-dependent bond weakening.³

In this work, we introduce ReactiveLJ, a phenomenological coordination-dependent LJ interaction that enables reversible bonding between coarse-grained stickers while retaining LJ-style two-body interactions. ReactiveLJ uses a smooth local coordination measure to reduce sticker attraction in crowded conditions and transitions smoothly between repulsive and attractive forces through force-continuous interpolation. Since interactions arise directly from the potential energy landscape, ReactiveLJ removes the need for predefined kinetic rules. As a result, the model can be used to study dynamics under non-equilibrium conditions.^{3,15}

We demonstrate ReactiveLJ across three distinct regimes. First, we study an associative single-chain system and quantify bond-swap frequency as a function of linker strength. Second, we simulate associative polymer melts across a range of cohesive strengths and characterize network self-assembly and gelation, chain mobility, linear viscoelastic relaxation, and microscopic exchange statistics. Third, we explore dilute-to-semidilute gelation under hydrodynamic conditions, using similar structural, dynamical, and kinetic measures, while also quantifying network formation rates as a function of sticker strength.

2 Models and Methods

2.1 Coarse-Grained Polymer Melts.

A standard Kremer–Grest bead–spring melt is combined with a coordination-dependent reactive interaction applied only between designated sticky monomers. All beads have unit mass $m = 1$, and reduced LJ units are used with $\sigma = 1$ and $\varepsilon = 1$. A melt consisting of $N_{\text{chains}} = 4000$ linear chains of length $N = 40$ is simulated in a periodic cubic box at monomer number density $\rho = 0.85$. A Langevin thermostat is used to maintain temperature $k_B T = 1.0$. Adjacent monomers are bonded by a Finitely Extensible Nonlinear Elastic (FENE) potential, with parameters $k = 30$ and $r_0 = 1.5$ and WCA parameters $\varepsilon = 1$ and $\sigma = 1$. Chain stiffness is imposed by a tabulated bending potential

$$U_{\text{bend}}(\theta) = \kappa(1 + \cos \theta), \quad \kappa = 1.5. \quad (1)$$

All backbone–backbone and backbone–sticker pairs are subject to the Weeks–Chandler–Andersen (WCA) potential to enforce excluded volume behavior. Sticker–sticker interactions are described exclusively by ReactiveLJ. Four stickers are placed deterministically and evenly along each chain backbone ($N = 40$) by replacing regular monomers with sticky monomers at indices $\{5, 15, 25, 35\}$ (zero-indexing).

2.2 Single-Chain Systems.

A single Kremer–Grest chain of length $N = 400$ is simulated in a large periodic cubic box of edge length $L = 100$ at $k_B T = 1.0$ using a Langevin thermostat. The same bonded and non-bonded interactions are employed as in the melt. Forty stickers are placed deterministically and evenly along the backbone at a spacing of 10 beads (indices $\{5, 15, 25, \dots, 395\}$, zero-indexed), preserving the same sticker fraction as the melt. An additional pre-production equilibration stage of equal length to the production run is performed before trajectory

collection.

2.3 Semi-Dilute Polymer Solutions.

Semi-dilute conditions are modeled by embedding Kremer–Grest chains of length $N = 40$ in a multi-particle collision dynamics¹⁹ (MPCD) solvent. The simulation box is set to match the melt reference state (4000 chains at $\rho = 0.85$) and snapped to the MPCD cell grid, giving a cubic box of edge length $L = 57$. The number of polymer chains is chosen to achieve a target polymer weight fraction of $w_p = 0.02$, yielding approximately 472 chains and $\sim 926,000$ MPCD solvent particles at a solvent number density of 5 particles per cell. Temperature is maintained at $k_B T = 1.0$ through stochastic rotation dynamics (SRD) collisions with a rotation angle of 130° and a collision period of 20 MD time steps. No thermostat is applied to the MD particles. All bonded and non-bonded interactions, as well as the placement of stickers, are identical to the melt.

2.4 ReactiveLJ Potential for Associative Bonds.

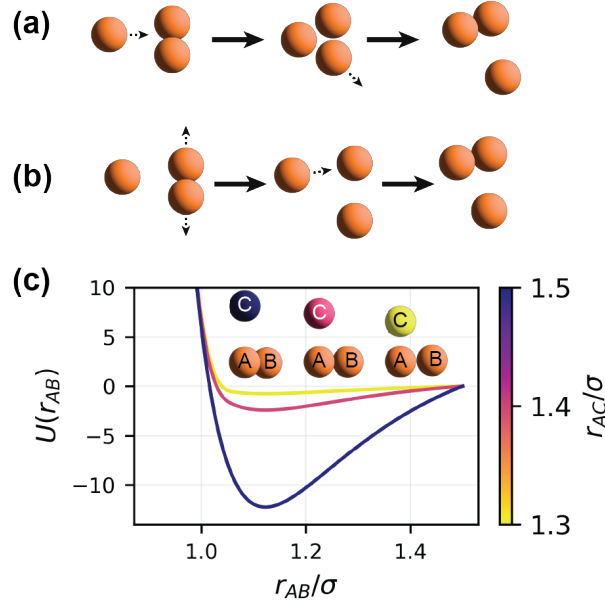


Figure 1: Schematic of coordination-dependent bond dynamics and the resulting ReactiveLJ interaction. (a) Associative mechanism: a third bead approaches an A–B pair and weakens the A–B attraction, enabling exchange of the partner. (b) Dissociative mechanism: the A–B pair first separates due to thermal fluctuations; a third bead then associates with one of the beads. (c) Crowding-dependent weakening implemented in ReactiveLJ. The pair potential $U(r_{AB})$ is progressively reduced as a third bead C approaches the A–B pair. Curve color indicates the A–C separation r_{AC} .

2.4.1 Overview

Associative bonding and exchange are controlled by a coordination-dependent Lennard–Jones interaction (ReactiveLJ) applied exclusively between beads of a designated reactive type $\mathcal{R}$ (stickers). For each reactive bead $i \in \mathcal{R}$, a smooth local coordination is computed as

$$C_i = \sum_{\substack{k \in \mathcal{R} \\ k \neq i}} w(r_{ik}), \quad (2)$$

where $w(r)$ is a compactly supported switching function on $[r_{\text{in}}, r_{\text{out}}]$,

$$w(r) = \begin{cases} 1, & r \leq r_{\text{in}}, \\ \frac{1}{2} \left[1 + \cos \left(\pi \frac{r - r_{\text{in}}}{r_{\text{out}} - r_{\text{in}}} \right) \right], & r_{\text{in}} < r < r_{\text{out}}, \\ 0, & r \geq r_{\text{out}}. \end{cases} \quad (3)$$

For each reactive pair (i, j) , a leave-one-out crowding variable is defined as

$$\text{raw}_{ij} = (C_i - w_{ij}) + (C_j - w_{ij}) = C_i + C_j - 2w_{ij}, \quad w_{ij} = w(r_{ij}), \quad (4)$$

and mapped to a smooth positive part

$$C_{ij}^{\text{exc}} = \frac{1}{2} \left(\text{raw}_{ij} + \sqrt{\text{raw}_{ij}^2 + \varepsilon_C^2} \right), \quad g_{ij} \equiv \frac{\partial C_{ij}^{\text{exc}}}{\partial \text{raw}_{ij}} = \frac{1}{2} \left(1 + \frac{\text{raw}_{ij}}{\sqrt{\text{raw}_{ij}^2 + \varepsilon_C^2}} \right), \quad (5)$$

with $\varepsilon_C = 10^{-6}$. Crowding reduces sticker cohesion through a weakening factor

$$W_{ij} = (1 + C_{ij}^{\text{exc}})^{-p}, \quad \frac{\partial W_{ij}}{\partial C_{ij}^{\text{exc}}} = -\frac{p W_{ij}}{1 + C_{ij}^{\text{exc}}}, \quad (6)$$

where p is the weakening exponent. A truncated-and-shifted LJ interaction is used as the underlying pair potential,

$$U_{\text{LJ}}(r) = 4\varepsilon \left[\left(\frac{\sigma}{r} \right)^{12} - \left(\frac{\sigma}{r} \right)^6 \right] - U_{\text{shift}}, \quad U_{\text{shift}} = 4\varepsilon \left[\left(\frac{\sigma}{r_{\text{cut}}} \right)^{12} - \left(\frac{\sigma}{r_{\text{cut}}} \right)^6 \right], \quad (7)$$

and, for $r < r_{\text{cut}}$, the ReactiveLJ construction corresponds to weakening only the attractive branch; in the absence of smoothing this reduces to

$$U_{ij}^{(0)}(r) = \max(U_{\text{LJ}}, 0) + W_{ij} \min(U_{\text{LJ}}, 0), \quad (8)$$

with $U_{ij} = 0$ for $r \geq r_{\text{cut}}$.

2.4.2 Force-continuous elbow smoothing.

To eliminate the force discontinuity at the repulsive–attractive transition, elbow smoothing is applied in a narrow window around the positive root r_{elbow} of $U_{\text{LJ}}(r) = 0$. Using

$$x_c = \left(\frac{\sigma}{r_{\text{cut}}} \right)^6, \quad x_* = \frac{1}{2} \left(1 + \sqrt{1 + 4(x_c^2 - x_c)} \right), \quad r_{\text{elbow}} = \sigma x_*^{-1/6}, \quad (9)$$

a crowding-dependent half-width is defined as

$$\delta_{ij} = \kappa_{\text{smooth}} \sigma (1 - W_{ij})^{\beta_{\text{smooth}}}, \quad r_1 = r_{\text{elbow}} - \delta_{ij}, \quad r_2 = r_{\text{elbow}} + \delta_{ij}, \quad (10)$$

with interpolation coordinate $t = \frac{r-r_1}{r_2-r_1}$. The elbow-smoothing window is restricted to the region where the shifted LJ interaction is evaluated by requiring $\sigma < r_{\text{in}} < r_{\text{out}} < r_{\text{cut}}$, so that the interpolation bounds remain within the cutoff. Within $r_1 < r < r_2$, the unsmoothed energy is replaced by a cubic Hermite interpolant,

$$U_{ij} = h_{00}U_1 + h_{10}L dU_1 + h_{01}U_2 + h_{11}L dU_2, \quad L = r_2 - r_1, \quad (11)$$

where

$$h_{00} = 2t^3 - 3t^2 + 1, \quad h_{10} = t^3 - 2t^2 + t, \quad h_{01} = -2t^3 + 3t^2, \quad h_{11} = t^3 - t^2, \quad (12)$$

and

$$U_1 = U_{\text{LJ}}(r_1), \quad dU_1 = \left. \frac{dU_{\text{LJ}}}{dr} \right|_{r_1}, \quad U_2 = W_{ij}U_{\text{LJ}}(r_2), \quad dU_2 = W_{ij} \left. \frac{dU_{\text{LJ}}}{dr} \right|_{r_2}. \quad (13)$$

Forces. Forces are obtained by differentiating the ReactiveLJ energy with respect to r_{ij} while accounting for the coordination dependence. The total pair force is decomposed as

$$\mathbf{F}_{ij} = \mathbf{F}_{ij}^{\text{pair}} + \mathbf{F}_{ij}^{\text{coord}}. \quad (14)$$

The pair term is defined as

$$\mathbf{F}_{ij}^{\text{pair}} = - \left. \frac{\partial U_{ij}}{\partial r_{ij}} \right|_{C_{ij}^{\text{exc}}} \hat{\mathbf{r}}_{ij}, \quad (15)$$

which reduces outside the elbow window to

$$\mathbf{F}_{ij}^{\text{pair}} = s_{ij} \frac{24\varepsilon}{r_{ij}} \left[2 \left(\frac{\sigma}{r_{ij}} \right)^{12} - \left(\frac{\sigma}{r_{ij}} \right)^6 \right] \hat{\mathbf{r}}_{ij}, \quad s_{ij} = \begin{cases} 1, & U_{\text{LJ}} > 0, \\ W_{ij}, & U_{\text{LJ}} \leq 0. \end{cases} \quad (16)$$

Adjoint variables are introduced as

$$\lambda_i = \frac{\partial E}{\partial C_i} = \sum_{j \neq i} g_{ij} \frac{\partial U_{ij}}{\partial C_{ij}^{\text{exc}}}, \quad (17)$$

with $\partial U_{ij}/\partial C_{ij}^{\text{exc}}$ evaluated analytically including the full chain rule through the Hermite-smoothed branch. Using

$$w'(r) = \begin{cases} -\frac{\pi}{2(r_{\text{out}} - r_{\text{in}})} \sin\left(\pi \frac{r - r_{\text{in}}}{r_{\text{out}} - r_{\text{in}}}\right), & r_{\text{in}} < r < r_{\text{out}}, \\ 0, & \text{otherwise,} \end{cases} \quad (18)$$

the coordination force is written as

$$\mathbf{F}_{ij}^{\text{coord}} = - \left[\lambda_i + \lambda_j - 2g_{ij} \frac{\partial U_{ij}}{\partial C_{ij}^{\text{exc}}} \right] w'(r_{ij}) \hat{\mathbf{r}}_{ij}. \quad (19)$$

2.4.3 Total energy and neighbor traversal.

The ReactiveLJ contribution to the total energy is

$$E = \sum_{\substack{i < j \\ i, j \in \mathcal{R} \\ r_{ij} < r_{\text{cut}}}} U_{ij}(r_{ij}, C_{ij}^{\text{exc}}), \quad (20)$$

and neighbor traversal is performed out to r_{cut} for coordination-force consistency.

Parameter choices. Unless otherwise stated, $\sigma_{\text{reactiveLJ}} = 1$, $p = 4$, elbow smoothing is enabled with $\kappa_{\text{smooth}} = 0.05$ and $\beta_{\text{smooth}} = 1$, $r_{\text{cut}} = 1.5\sigma$, $r_{\text{in}} = 1.3\sigma$, and $r_{\text{out}} = 1.5\sigma$. Sticker cohesive strength is varied by selecting $\varepsilon_{\text{reactiveLJ}} \in \{3, 6, 9, 12, 15, 18\}$.

2.5 Simulation Protocol.

All simulations are performed in reduced Lennard–Jones units with bead mass $m = 1$, characteristic length $\sigma = 1$ (set by the LJ diameter), and characteristic energy $\varepsilon = 1$, which together define the simulation time unit $\tau = (m\sigma^2/\varepsilon)^{\frac{1}{2}}$. Initial chain conformations are generated by random-walk placement with rejection sampling using an initial bond length of 0.97 and a minimum nonbonded placement distance of 0.8. Velocities are initialized from a Maxwell–Boltzmann distribution at $k_{\text{B}}T = 1.0$.

Dynamics are integrated with Langevin (NVT) equations of motion using timestep $\Delta t = 0.005\tau$ and damping time $\tau_T = 100\tau$ unless otherwise specified. Center-of-mass drift introduced by Langevin forcing is removed every 100 steps by subtracting the instantaneous center-of-mass velocity. A neighbor-list buffer of 0.4 is used.

Each replicate follows the schedule:

1. **Unsticky equilibration:** 10^5 steps are run with stickers present but without ReactiveLJ attraction.
2. **ReactiveLJ activation:** the timestep is reduced to $\Delta t = 10^{-5}\tau$, reactive types are

assigned to the predetermined sticker beads, and ReactiveLJ interactions are enabled.

3. **Reactive equilibration:** 10^6 steps are run while $\varepsilon_{\text{reactiveLJ}}$ is ramped from 10^{-4} to the target value in 1000-step segments, followed by a timestep ramp from $10^{-5}\tau$ to 0.005τ over 10^5 steps.
4. **Production:** 10^6 steps are run at the target $\varepsilon_{\text{reactiveLJ}}$.

Trajectory frames are written every 10^4 steps and only during production. For each $\varepsilon_{\text{reactiveLJ}}$, 10 independent replicates are performed, yielding 60 production trajectories in total.

2.6 Energy-Conservation Validation.

Energy-conservation validation runs are performed at $\varepsilon_{\text{reactiveLJ}} = 3$ and 18 (10 replicates each) by switching to NVE dynamics during production and logging the sum of kinetic and potential energy. To reduce numerical integration error and associated energy drift, a smaller timestep of $\Delta t = 5 \times 10^{-4}$ is used. Each NVE production run is continued for 10^7 steps.

2.7 Molecular Analysis.

Network self-assembly and gelation are characterized by identifying chain clusters connected by associative bonds. Two chains are defined as connected if any sticker–sticker pair between them satisfies $r < r_{\text{bond}}$. In this analysis, the bond criterion is set to the Lennard–Jones inflection distance,

$$r_{\text{bond}} = r_{\text{thresh}} = \sigma \left(\frac{26}{7} \right)^{1/6}. \quad (21)$$

Connected components of the resulting chain graph define cluster sizes, from which the chain cluster-size distribution is computed. The mean-field degree of gelation $\mathcal{E}$ is calculated from the equilibrium probability for a sticker to be unpaired, p_{open} , by defining the bonding probability $p = 1 - p_{\text{open}}$ and the critical bonding probability $p_c = \frac{1}{f-1}$, and then evaluating $\mathcal{E} = \frac{p-p_c}{p_c}$.

Monomer mobility is measured by the mean-squared displacement

$$g_1(t) = \langle |\mathbf{r}_i(t) - \mathbf{r}_i(0)|^2 \rangle, \quad (22)$$

averaged over all monomers and time origins.

Linear viscoelasticity is obtained from equilibrium stress autocorrelations through the Green–Kubo relation

$$G(t_\ell) = \frac{V}{k_B T} \langle \delta\sigma_{\alpha\beta}(t_0 + t_\ell) \delta\sigma_{\alpha\beta}(t_0) \rangle_{t_0}, \quad \alpha\beta \in \{xy, xz, yz\},$$

with averaging over the three off-diagonal shear components. Here $G(t_\ell)$ is the stress relaxation modulus, V is the simulation box volume, $k_B T$ is the thermal energy, $\delta\sigma_{\alpha\beta} = \sigma_{\alpha\beta} - \langle \sigma_{\alpha\beta} \rangle$, and $\langle \cdot \rangle_{t_0}$ denotes averaging over time origins. The lag time is discretized as $t_\ell = \ell \Delta t_{\text{frame}}$, and the maximum lag is fixed to $t_{\text{max}} = 1250\tau$ to span the regime used for exponential decay fits.

Microscopic kinetics are quantified from Boolean bond histories constructed using the same distance criterion $r < r_{\text{bond}}$. Bond and unpaired-site lifetimes are extracted from exponential decays of the corresponding time-correlation functions. Exchange events are classified by comparing partners in adjacent saved frames: a bond formed at frame t is labeled associative-type if the forming sticker has a different partner at frame $t - 1$, and dissociative-type otherwise. Rates of associative-type and dissociative-type events are reported per unpaired sticker population.

3 Results and discussion

3.1 Single-chain systems

We first consider a single-chain model under Langevin dynamics, simulating sticker bonding and unbonding across a range of linker stickiness values $\varepsilon_{\text{reactiveLJ}} \in \{6, 12, 15, 18\}$. For

each condition, we compute the average bond persistence time, the associative and dissociative bond formation rates, and the radius of gyration R_g , enabling a comparison between associative bond kinetics and chain morphology.

3.1.1 Bond persistence

3.2 Melt systems

We next consider polymer melts to examine how viscoelastic behavior depends on sticker interaction strength. Simulations are performed as described in Section 2.1.

3.2.1 Bond persistence and brachiation time

As an initial characterization of the reactive melt model, we examined how sticker persistence depends on the reactive attraction strength, $\varepsilon_{\text{reactiveLJ}}$. As described in Section 2.7, we identified sticker–sticker bonds in each saved configuration and constructed the bond-lifetime correlation function

$$f(t) = \langle s_i(0)s_i(t) \rangle_i, \quad (23)$$

where $s_i(t) = 1$ when bond i is present and $s_i(t) = 0$ otherwise. We extracted the characteristic sticker persistence time, τ_s , by fitting the initial semilog-linear portion of the decay to a single exponential,

$$f(t) \sim \exp\left(-\frac{t}{\tau_s}\right). \quad (24)$$

Figure 2a shows that increasing $\varepsilon_{\text{reactiveLJ}}$ shifts the decay of $f(t)$ to longer times, which indicates progressively longer-lived sticker associations. Stronger attractions increase both the energetic cost of dissociation and the coordination-dependent penalty associated with nearby reactive beads, so they suppress the multi-body encounters that would otherwise promote bond reorganization. In our parameterization with weakening exponent $p = 4$, the decays for $\varepsilon_{\text{reactiveLJ}} \in \{6, 12\}$ occur quickly, whereas substantially slower relaxation emerges at larger $\varepsilon_{\text{reactiveLJ}}$. We therefore focus the dump-interval calibration in Figure 2b

on $\varepsilon_{\text{reactiveLJ}} = 12, 15$, and 18 , where sticker dynamics are sufficiently persistent to be most relevant for associative melt behavior.

Liu and O’Connor³ showed that the measured sticker persistence time and brachiation time can both depend on the interval Δt between recorded configurations (“dump interval”). To test whether our sampling interval was adequate, we recomputed τ_s and the brachiation time τ_b over effective dump intervals spanning 500 to $10,000 \tau_{\text{LJ}}$ by subsampling the saved trajectories. We fit τ_s by applying a single-exponential fit to the initial semilog-linear portion of the bond-lifetime correlation decay. Likewise, we fit τ_b in the same manner, but using the autocorrelation of the indicator for a sticker remaining unpaired,

$$g(t) = \langle o_i(0)o_i(t) \rangle_i, \quad (25)$$

where $o_i(t) = 1$ when sticker i is unpaired and $o_i(t) = 0$ otherwise. Figure 2b shows that τ_s varies only weakly across this range of dump intervals, whereas τ_b changes appreciably with Δt . Following the half-lifetime criterion of Liu and O’Connor, the reference line $y = 2\Delta t$ marks the condition that the measured relaxation time exceeds twice the sampling interval. In other words, we record configurations at least twice within one characteristic decay time when a data point lies above this line. Meeting this criterion reduces the likelihood that bond-breaking, recombination, or rebinding events occur between successive saved frames without being resolved. At the production sampling interval, $\Delta t = 500 \tau_{\text{LJ}}$, all measured τ_s values remain above the reference line, whereas the corresponding τ_b values do not do so consistently. We therefore conclude that our sampling interval resolves the sticker persistence time τ_s , but it does not adequately resolve the brachiation time τ_b ; accordingly, we focus the subsequent analysis on τ_s only.

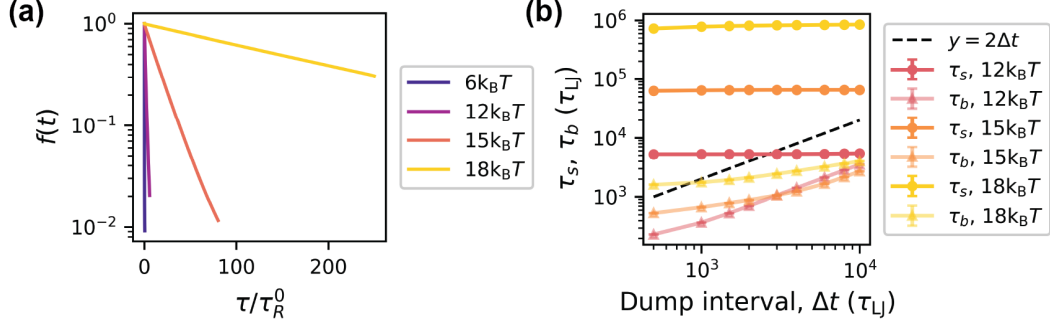


Figure 2: Bond-persistence dynamics and dump-interval calibration for the associative melt. (a) Bond-lifetime autocorrelation, $f(t)$, is plotted against normalized lag time, τ/τ_R^0 , for reactive-LJ attraction strengths $\epsilon_{\text{reactiveLJ}} \in \{6, 12, 15, 18\} k_B T$. The initial semilog-linear region of each curve is shown. (b) Sticker persistence times τ_s (circles) and brachiation times τ_b (triangles) are obtained from fitting a single exponential decay to the initial semilog-linear portions of the bond-lifetime and open-sticker correlation decays. These times are plotted against effective dump interval Δt for $\epsilon_{\text{reactiveLJ}} \in \{12, 15, 18\} k_B T$, and error bars represent standard errors across replicates. Error bars are generally smaller than the symbol size. The dashed black line indicates $y = 2\Delta t$.

3.2.2 Gel point and network formation

The mean-field degree of gelation $\mathcal{E}$, as described by theory, is a measure of how close a network structure is to the point at which connectivity becomes system-spanning; this point is referred to as the “critically percolated gel structure”. In other words, at this critical structure, it becomes possible to reach any chain in the network by following a continuous path of connected chains. Mathematically, $\mathcal{E}$ is described as

$$\mathcal{E} = \frac{p - p_c}{p_c} = \frac{(1 - p_{\text{open}}) - p_c}{p_c} \quad (26)$$

where p is the probability of a sticky site being bonded and $p_c = \frac{1}{f-1}$ is the critical gelation bonding probability where f is the number of stickers per chain. Figure!3a shows the probability $P(M)$ of a chain being part of a cluster of size M , and we observe that for the default case where linkers are simply hard spheres ($\epsilon_{\text{reactiveLJ}} = 0$), $P(M)$ decays according to a power-law relationship; that is, the probability of a chain being part of a network of size M decreases according to the power-law relationship $P(M) \sim M^{-2.3}$. This result is

approximately consistent with mean-field gelation theory, which states that $P(M) \sim M^{-2.5}$.
²⁰ [FACT CHECK THE PREVIOUS STATEMENT PLEASE]. As stickiness increases for $\varepsilon_{\text{reactiveLJ}} > 0$, what we observe is that very quickly, clusters begin to span nearly the whole network of 4000 chains, hence we observe a near-vertical stack of points at $M \approx 4000$. What this tells us is that in the presence of linker associations, the chains very quickly begin to form a network, which is not surprising given the number density of our box at 0.85. [FACT CHECK THE PREVIOUS CLAIM PLEASE].

To quantify network formation, we compute the degree of gelation $\mathcal{E}$, as defined in Section 2.7. Here, full saturation corresponds to $\mathcal{E} = 2$, while full gelation at $\mathcal{E} = 1$ corresponds to the point in which all chains in the network are connected [FACT CHECK THE PREVIOUS STATEMENT PLEASE]. Figure 3b, shows that $\mathcal{E}$ increases toward this limit with increasing $\varepsilon_{\text{reactiveLJ}}$. This behavior suggests that as linker stickiness increases, the melt becomes more likely to form a network [FACT CHECK THE PREVIOUS STATEMENT PLEASE].

Network “defects” arising from intramolecular bond formation are shown in Figure S1a; the intramolecular bond fraction increases with $\varepsilon_{\text{reactiveLJ}}$ but remains near the $\sim 5\%$ level reported by Liu and O’Connor. As a result

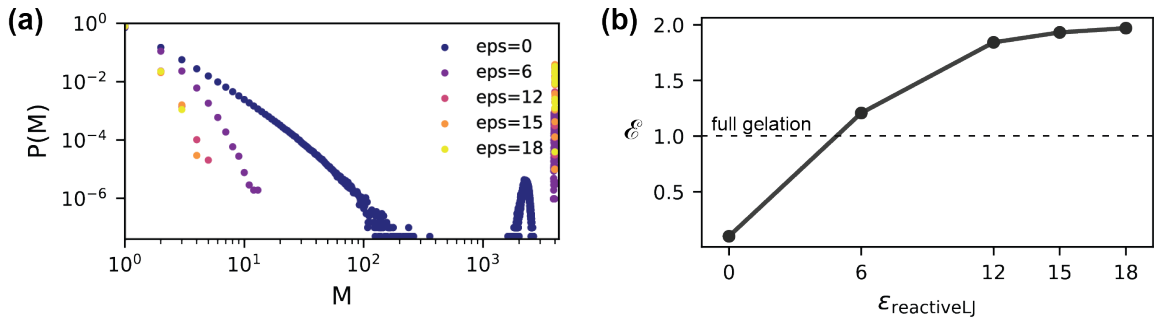


Figure 3: (a) Cluster size distribution for various linker stickinesses and (b) mean degree of gelation averaged over all production frames for each value of $\varepsilon_{\text{reactiveLJ}}$.

3.2.3 Viscoelastic properties

To characterize the equilibrium viscoelastic response of the melt networks, we analyze the virial-tensor time series and the monomer trajectories. The network relaxation modulus, $G(\tau)$, is obtained from a Green–Kubo stress autocorrelation constructed from the rotationally averaged deviatoric stress, using the three shear components and the three normal-stress differences of the virial tensor. Monomer transport is quantified through the monomer mean squared displacement (MSD). In both cases, lag time is normalized by the Rouse relaxation time of the non-associating reference melt, τ_R^0 , following Liu and O’Connor.³ Figure 4 shows representative results for $\varepsilon = 0, 6, 12, 15$, and 18.

At short lag times, $G(\tau)$ decays approximately as a power law, consistent with Rouse-like relaxation and indicating that the earliest-time response is only weakly affected by association. As linker stickiness increases, stress relaxation becomes progressively slower and develops a transient plateau at intermediate lag times for the stronger linkers, reflecting the emergence of transient network elasticity. Figure 4b shows the corresponding MSD curves, plotted as weighted mean values across replicates at each lag for each sticker strength. In the same intermediate-time regime, the MSD exhibits increasingly subdiffusive growth as ε increases, with slopes well below that expected for free diffusion. At later times, the MSD curves steepen relative to this intermediate-time regime, but this crossover is shifted to progressively later times as linker stickiness increases. Taken together, these trends indicate that increasing linker stickiness strengthens transient elastic response while suppressing chain mobility, in qualitative agreement with the equilibrium viscoelastic and diffusive behavior reported for associative polymer melts by Liu and O’Connor.³

3.2.4 Equilibrium network kinetics

Based on the melt simulation trajectories, we compute the bond persistence time and the brachiation time as functions of linker interaction strength $\varepsilon_{\text{reactive}}$, as defined in Section 2.7. Both timescales increase exponentially with $\varepsilon_{\text{reactive}}$, as shown in Figure 2a, consistent with

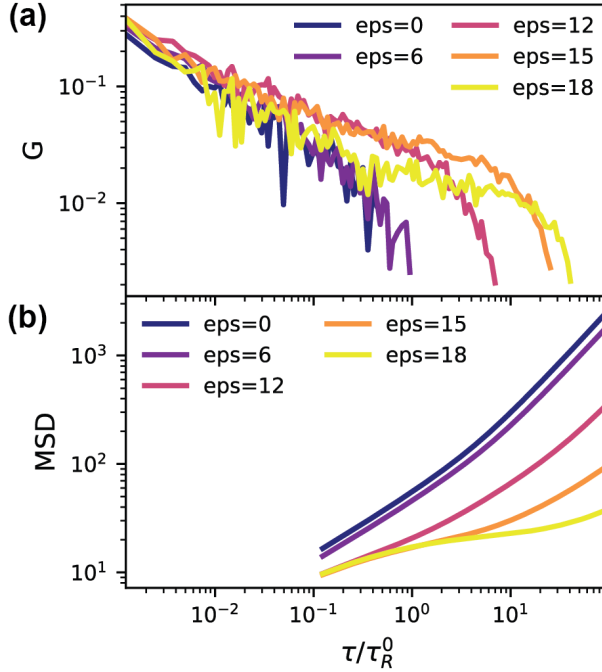


Figure 4: (a) Network relaxation modulus, G , and (b) monomer mean squared displacement (MSD) as functions of normalized time, τ/τ_R^0 , for sticker strengths $\epsilon_{\text{reactiveLJ}} = 0, 6, 12, 15$, and 18 .

the single-chain results. We additionally quantify the associative and dissociative exchange rates, shown in Figure 2b, which are maximal at intermediate $\epsilon_{\text{reactive}}$. At low $\epsilon_{\text{reactive}}$, stickers rarely remain bound long enough to sustain an associative network, whereas at high $\epsilon_{\text{reactive}}$, bond breaking and partner exchange are suppressed by the stability of the bonded pair and by the coordination-dependent repulsion that disfavors a third bead approaching a bonded pair. These results indicate that linker strength controls not only the lifetime of individual bonds but also the rate at which the network can reorganize through partner exchange.

3.3 Semi-dilute solutions

Third, we examine semidilute polymer networks, using the simulation protocol detailed in Section 2.3 and analyzing the same set of network-formation observables as in the melt case (Section ??). Given that most polymer chemistries and processing pathways occur in solution, this regime holds the most experimental relevance. Figure 6a illustrates that increasing

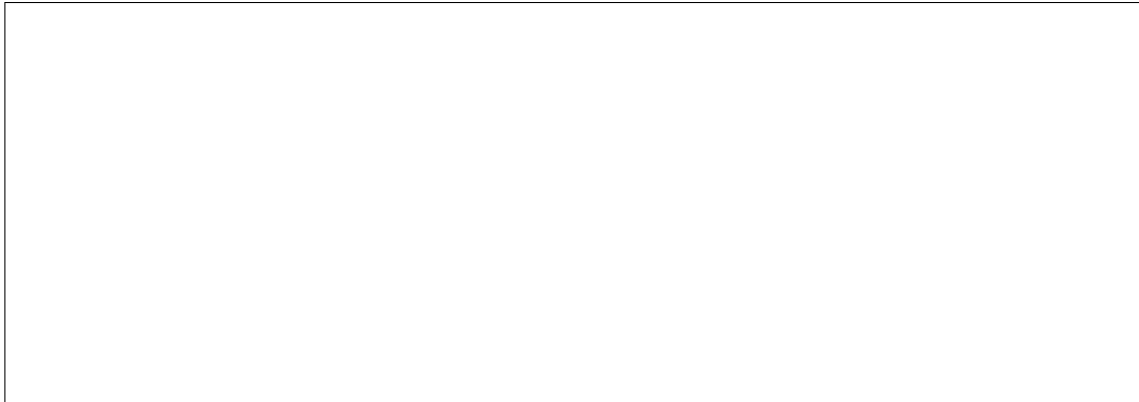


Figure 5: Placeholder for melt bond persistence and exchange kinetics figure.

linker stickiness results in longer bond persistence times and brachiation times, aligning with both the single-chain and melt results. These trends are also consistent with existing literature. Sticky Rouse theory predicts that the bare sticker lifetime increases exponentially with association energy, $\tau_b \sim \exp(\varepsilon/k_B T)$, and demonstrates that correlation effects further enhance this dependence through bond lifetime renormalization. Consequently, stronger stickers extend bond persistence times and chain-level relaxation timescales,^{21,22} with corresponding effects on bulk network rheology,²³ consistent with transient network experiments and supramolecular network studies.^{24–26} Overall, our simulations phenomenologically capture experimental trends indicating that linker strength impacts bond persistence times and network relaxation behavior in solution-phase associative polymer networks.

3.4 Computational efficiency comparison

We compare the achievable simulation throughput (in reduced time units per wall-clock time) for systems governed by ReactiveLJ and by an analogous Tersoff-style bond-order potential.¹⁷ The Tersoff-analog fitting protocol is described in SectionXX. FigureYY shows that ReactiveLJ attains slightly, but statistically significantly, higher throughput than the Tersoff analog across the range of linker stickiness values examined.

This improvement in performance is likely due to the simpler nature of the ReactiveLJ model compared to the Tersoff analog. ReactiveLJ is formulated in terms of pair interactions

modulated by a coordination-dependent weakening factor, whereas Tersoff-type potentials incorporate explicit bond-order and angular dependencies, which introduce additional many-body arithmetic for each interacting neighborhood.¹⁷ Throughput is sensitive to the effective neighbor list workload because the pair-force computation time increases when the neighbor list contains more candidate interactions. Additionally, as an LJ-based potential, ReactiveLJ exhibits harder repulsive cores than Morse-style potential such as Tersoff. This can reduce crowding and further streamline execution, as neighbor-list construction itself can become more expensive when local neighbor environments are large.

ReactiveLJ is complementary to the existing RevCross bond-swap approaches.¹⁶ In the HOOMD-blue implementation described by Ciarella and Ellenbroek, the swap potential is constructed on top of an attractive interaction defined specifically between particles of type A and type B,¹⁶ so at least two distinct species are required by construction. By contrast, many experimentally relevant associative systems involve indistinguishable reactive moieties that self-associate into reversible dimers or junctions without requiring a distinct crosslinker species, including telechelic polymers with identical attractive end groups and supramolecular polymers based on self-complementary ureidopyrimidinone (UPy) hydrogen-bonding motifs.^{27–29} In such settings, ReactiveLJ provides a fast, phenomenological bond-order-like interaction that naturally supports same-type reversible association while avoiding explicit angular dependence.

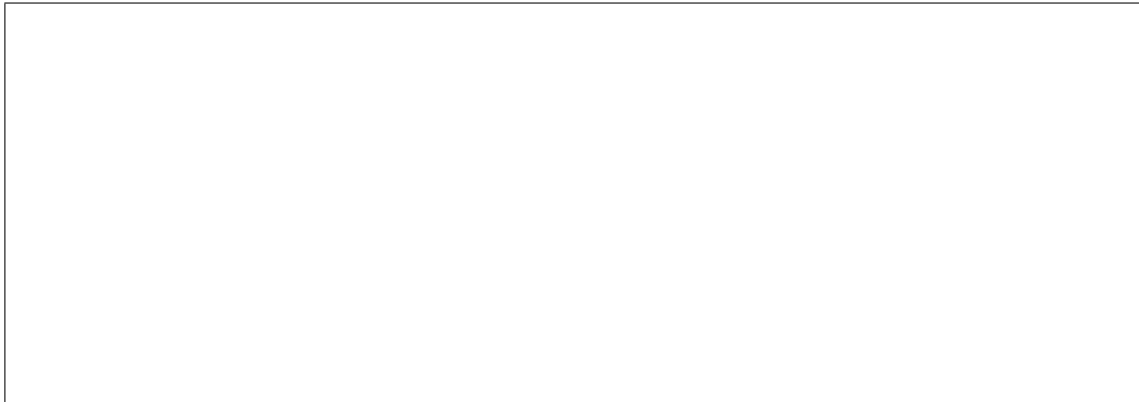


Figure 6: Placeholder for melt network formation and dynamics figure.

4 Conclusions

This study introduces ReactiveLJ, a crowding-dependent reactive potential designed to model reversible associations in coarse-grained polymer systems. By adjusting the stickiness of interactions between reactive beads through the parameter $\varepsilon_{\text{reactiveLJ}}$, bond formation and breakage dynamics can be controlled without the need for distinguishable reactive species.

In single-chain systems under Langevin dynamics, ReactiveLJ accurately reproduces the expected exponential dependence of bond persistence time on interaction strength. The coordination-dependent repulsion suppresses multi-body encounters and provides a physically motivated mechanism for bond selectivity. In coarse-grained melts, both the degree of gelation and the bead mean-squared displacement vary with $\varepsilon_{\text{reactiveLJ}}$, indicating that linker strength governs the thermodynamic extent of network formation and the dynamic accessibility of network configurations. In semidilute solution, coupling ReactiveLJ with MPCD preserves these trends under hydrodynamic conditions, with bond persistence times and brachiation times remaining consistent with sticky Rouse theory and supramolecular network experiments.

Compared to the existing Tersoff potential, ReactiveLJ requires fewer parameters and incurs lower computational cost. Additionally, it offers greater flexibility compared to the LJ-based RevCross bond-swap potential, as it does not necessitate the presence of two chemically distinguishable species for bond swaps. These properties collectively suggest that ReactiveLJ can serve as a versatile coarse-grained reactive potential for associative polymer systems involving indistinguishable reactive species. Future research should explore ReactiveLJ to investigate structure–property relationships in reversibly cross-linked single-chain nanoparticle systems, network healing kinetics in transient network hydrogels, and the response of end-linking associating networks to flow. This will broaden its applicability to the design of stimuli-responsive and self-healing polymeric materials.

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Supporting Information Available

This will usually read something like: “Experimental procedures and characterization data for all new compounds. The class will automatically add a sentence pointing to the information on-line:

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